Automated Traffic Anomaly Localization and Signature-Based Threat Mitigation Engine

Parva Kothari
23BIT0053
parva.kothari2023@vitstudent.ac.in
SCORE, VIT, Vellore

K. Jaya Vishal
23BIT0197
kalimisetty.jaya2023@vitstudent.ac.in
SCORE, VIT, Vellore

A. Teja
23BIT0296
akurathi.teja2023@vitstudent.ac.in
SCORE, VIT, Vellore

Abstract

Network intrusion detection systems (NIDS) must accurately identify a wide spectrum of attack types in the presence of severe class imbalance, where normal traffic vastly outnumbers attack samples. Conventional deep neural networks trained under these conditions exhibit strong bias toward majority classes, systematically under-detecting rare but critical attack categories. In this paper, we propose a soft-gated Mixture-of-Experts (MoE) neural network for multiclass network intrusion classification on the UNSW-NB15 benchmark dataset.

Our architecture comprises three structurally heterogeneous expert sub-networks — a medium multi-layer perceptron (MLP), a deep MLP, and a wide MLP with dropout regularization — combined via a learned softmax gating network that routes each sample to a weighted combination of experts based on its input features. To address class imbalance without synthetic data augmentation, we introduce a square-root balanced class-weighting scheme that softens extreme penalty ratios and improves generalization on minority classes. Training is stabilized through cosine annealing learning rate scheduling, gradient norm clipping, and early stopping governed by macro-averaged F1-score rather than accuracy. Evaluated on a held-out test set (15% stratified split), the proposed MoE model produces competitive multiclass classification performance and demonstrates measurable improvement over single-network baselines in macro F1.

The dual evaluation strategy — reporting both full multiclass and binary (Normal vs. Attack) metrics — provides practitioners with an operationally relevant view of system behavior. Our results affirm that heterogeneous expert specialization, imbalance-aware loss weighting, and F1-driven model selection collectively yield a more robust NIDS than standard deep classifiers.

Keywords: Mixture of Experts, Intrusion Detection, UNSW-NB15, Neural Networks, Ensemble Learning, Cyber Threat Mitigation.

1. Introduction

The proliferation of connected devices and increasingly sophisticated cyber threats has made network intrusion detection a critical challenge in modern cybersecurity. The rapid proliferation of technologies such as IoT, cloud computing, and smart devices has led to massive growth in annual network traffic, making robust intrusion detection essential for protecting sensitive data. Traditional signature-based systems are insufficient for detecting novel attacks, driving demand for machine learning and deep learning approaches.

Anomaly-based NIDS, which learn the statistical profile of normal and malicious traffic, have shown substantial promise. However, two fundamental challenges impede their deployment: (i) class imbalance, where certain attack categories represent a fraction of a percent of total samples, and (ii) attack diversity, where a single homogeneous model must simultaneously master the discrimination boundaries of nine structurally dissimilar attack families (Backdoor, DoS, Exploits, Fuzzers, Generic, Reconnaissance, Shellcode, Worms, and Normal traffic).

Ensemble methods and multi-model architectures have been proposed to address these challenges. Unlike traditional IDS that rely on static rules or isolated classifiers, hybrid models leverage the combined strengths of multiple learning approaches within a unified architecture. However, most such approaches either rely on hard voting between independently trained models or fail to incorporate input-conditional routing — that is, they do not adapt which sub-model is most trusted based on the characteristics of each incoming sample.

The Mixture-of-Experts (MoE) paradigm offers a principled solution. MoE is a machine learning technique where multiple expert networks are used to divide a problem space into homogeneous regions, representing a form of ensemble learning. Unlike static ensembles, MoE employs a gating function that produces a sample-specific combination of expert outputs, allowing the model to implicitly specialize different experts for different sub-problems within the same task.

We apply this paradigm to multiclass NIDS on UNSW-NB15 for the first time in a tabular, non-image-converted setting, and make the following contributions:

1. We design a heterogeneous 3-expert MoE with a learned softmax gating network for multiclass network intrusion classification, where each expert has a distinct depth-and-dropout profile to encourage structural diversity.
2. We propose a square-root balanced class-weighting strategy that smooths the penalty ratio for extreme imbalances without resampling, preserving the original class distribution.
3. We adopt macro F1-score as the early stopping criterion, arguing that it is statistically superior to accuracy for imbalanced multiclass NIDS evaluation.
4. We conduct dual evaluation at both the multiclass and binary (Normal vs. Attack) levels, aligning technical metrics with operational security requirements.

2. Related Work

2.1 The UNSW-NB15 Dataset

The UNSW-NB15 dataset contains a hybrid of real modern normal traffic and contemporary synthesized attack activities. It was created to address limitations of older benchmarks like KDD98 and NSLKDD, which do not reflect modern network traffic or low-footprint attacks. It has since become the de facto benchmark for evaluating novel NIDS approaches. Studies have consistently shown that UNSW-NB15 is more complex than KDD99 and is considered a new benchmark dataset for evaluating NIDS.

2.2 Machine Learning and Deep Learning for NIDS

A broad spectrum of ML classifiers have been applied to UNSW-NB15. Machine learning algorithms have achieved 98.6% accuracy on UNSW-NB15 in binary classification, with Random Forest and Logistic Regression performing best in that dataset, while deep learning models such as MLP and LSTM excelled in other settings. Deep learning approaches have progressively surpassed classical ML. Existing machine learning-based NIDS often require complex feature engineering and domain expertise, while deep learning approaches, though powerful, struggle with imbalanced data, resulting in bias toward normal traffic and reduced effectiveness for detecting rare attacks.

Hybrid architectures have shown promise. A hybrid deep learning model combining Bidirectional LSTM, self-attention feature weights, and Random Forest classification achieved binary and multiclass classification accuracies of 99.56% and 99.45% respectively on UNSW-NB15.

2.3 Class Imbalance in NIDS

Class imbalance is a central and persistent challenge in intrusion detection benchmarks. Benchmark datasets in intrusion detection simulate real-network traffic by including more normal traffic than attack samples, causing the training data to be imbalanced and creating difficulties for detecting certain types of attacks. Common mitigation strategies include SMOTE-based oversampling, undersampling, and cost-sensitive learning. Class imbalance directly affects the ability of machine learning models to identify minority-class threats, and the judicious integration of resampling with machine learning models can create more robust and adaptive intrusion detection systems. Cost-sensitive training with class weighting avoids generating synthetic samples and therefore introduces no risk of overfitting to artificially constructed minority instances.

2.4 Ensemble and Multi-Model Approaches

Ensemble methods have demonstrated consistent performance gains on NIDS benchmarks. An ensemble classifier integrating Balanced Bagging, XGBoost, and Random Forest with Hellinger Distance Decision Trees, augmented with novel threshold-adjustment algorithms, can address both data imbalance and class overlap in intrusion detection. Deep learning ensembles extend this idea to neural architectures. A lightweight ensemble deep learning model combining LSTM, GRU, and DNN with XGBoost-based feature selection and SMOTE achieved 99.4% accuracy on the RT-IoT2022 dataset, demonstrating that ensemble deep learning is effective for imbalanced IoT intrusion detection.

2.5 Mixture-of-Experts Architectures

The MoE paradigm has evolved into a diverse architectural space, with variants differing in scale, number of experts, and routing strategies. Early conceptual foundations established that a gating function routes each input to the most relevant expert, enabling modularity and competitive learning. In dense MoE, each expert network accepts the same input and generates an output, while a gating network — typically a linear-softmax network — combines their outputs.

Critically, MoE has only recently been explored for network intrusion detection. The integration of MoE layers into a deep neural network for intrusion detection in 5G networks represents a novel direction, where models conditioned on the input offer benefits including efficiency, expressive power, adaptiveness, and compatibility because they selectively activate their components. Our work differs by applying a fully dense, heterogeneous MoE directly to tabular UNSW-NB15 features without image conversion, and by focusing on the nine-class multiclass problem under realistic class imbalance.

3. Dataset and Preprocessing

3.1 UNSW-NB15

The UNSW-NB15 dataset contains network traffic records with 49 raw features covering flow statistics, protocol attributes, connection characteristics, and content-based features. The target label attack_cat specifies one of ten categories: Normal, Backdoor, DoS, Exploits, Fuzzers, Generic, Reconnaissance, Shellcode, and Worms.

3.2 Preprocessing Pipeline

- Feature loading: Column names are loaded from the companion NUSW-NB15_features.csv file and applied to the raw CSVs, which carry no header. Non-informative identifiers (ID, source IP, destination IP) are dropped to prevent data leakage.
- Label normalization: The attack_cat field contains inconsistent casing and whitespace (e.g., "Backdoors", "backdoor", " Normal "). All variants are mapped to a canonical set of nine classes. Null or empty entries are assigned to the Normal class.
- Missing value imputation: Numeric missing values are filled with column medians; categorical missing values with column modes. Infinite values are replaced with NaN and subsequently imputed.
- Categorical encoding: The three nominal features (protocol, service, connection state) are one-hot encoded, yielding approximately 196 total features after expansion.
- Data splitting: The dataset is split 70/15/15 into train, validation, and test sets using stratified sampling to preserve class proportions at each split.
- Feature scaling: StandardScaler is fit exclusively on the training set and applied to validation and test sets, ensuring no information leakage. Remaining hex-encoded port columns are coerced to integer representation before scaling.

3.3 Class Imbalance Characterization

UNSW-NB15 is severely imbalanced. Generic and Exploits attacks account for the vast majority of attack-type records, while Shellcode and Worms represent fewer than 0.5% of total samples each. Balanced class weights computed naively produce extreme ratios (sometimes 50:1 or higher), which destabilize training. We address this in Section 4.3.

4. Proposed Methodology

4.1 Architecture Overview

The proposed model is a dense soft-gated Mixture-of-Experts network comprising three structurally distinct expert sub-networks and a lightweight gating network.
Given an input vector $x \in \mathbb{R}^d$, the model computes:

$$\hat{\mathbf{y}} = \boldsymbol{\Sigma}_i\, \mathbf{g}_i(\mathbf{x}) \cdot \mathbf{f}_i(\mathbf{x})$$

where $g(x) = \text{softmax}(W_g \cdot \text{ReLU}(U_g \cdot x))$ are the scalar gate weights and $f_i(x)$ is the logit vector produced by the i-th expert.

4.2 Expert Architectures

Three experts with deliberately varied structural profiles are used to encourage diversity:

- Expert 1 (Medium MLP): Linear(d→256) → BN → ReLU → Linear(256→128) → BN → ReLU → Linear(128→C). Captures mid-complexity feature interactions with compact representation.
- Expert 2 (Deep MLP): Linear(d→512) → BN → ReLU → Linear(512→256) → BN → ReLU → Linear(256→128) → BN → ReLU → Linear(128→C). Maximizes representational depth for complex, non-linear attack patterns.
- Expert 3 (Wide + Dropout MLP): Same topology as Expert 2 but with Dropout(p=0.4) after each BatchNorm. Acts as a regularized generalist, preventing over-concentration of gradients during imbalanced training.

where BN denotes Batch Normalization, C is the number of classes (9), and d is the input feature dimension.

4.3 Gating Network

The gating network is a two-layer MLP: Linear(d→64) → ReLU → Linear(64→3), followed by Softmax. It operates on the same raw input features as the experts and learns to continuously weight expert contributions per sample during training. No sparsity constraint is imposed — all three experts contribute to every prediction, weighted by their gate scores.

4.4 Square-Root Balanced Class Weighting

To handle class imbalance without synthetic augmentation, we compute per-class loss weights as:

$$\mathbf{w}_i = \mathbf{clip}(\sqrt{(\mathbf{w}_{\text{balanced},i} / \mathbf{mean}(\mathbf{w}_{\text{balanced}}))}, 0.1, 5.0)$$

where $w_{\text{balanced},i}$ is the standard sklearn balanced weight for class i. The square-root transformation softens extreme ratios, and clipping to [0.1, 5.0] prevents gradient instability for the rarest classes. These weights are applied to the CrossEntropyLoss criterion at training time.

4.5 Training Configuration

Hyperparameter	Value
Optimizer	Adam
Learning Rate (initial)	3×10^{-4}
LR Scheduler	CosineAnnealingLR ($T_{max} = 80$)
Min LR	1×10^{-6}
Weight Decay	1×10^{-4}
Batch Size	512
Max Epochs	80
Early Stopping Patience	12 epochs
Early Stopping Metric	Macro F1 (validation)
Gradient Clip Norm	5.0
Dropout (Expert 3)	0.4

Table 1: Training Configuration

CosineAnnealingLR is chosen over step-decay schedules because it smoothly reduces the learning rate without abrupt drops that can destabilize minority-class gradient signals near decision boundaries.

4.6 Early Stopping on Macro F1

Early stopping is triggered when validation macro-averaged F1 fails to improve for 12 consecutive epochs. We specifically choose macro F1 over accuracy because accuracy in a severely imbalanced dataset can remain high while rare-class detection collapses — a misleading signal for model selection. Macro F1 equally weights all classes regardless of sample count, providing an honest stopping criterion for the NIDS multiclass problem.

5. Results

The evaluation of the Main Neural MoE model yields profound insights into the system's operational dynamics. The findings are evaluated at both the multiclass anomaly localization level and a simplified binary threat detection level, providing a comprehensive assessment of the model's capabilities. Table 2 summarizes the core final trained scores of the model on the validation and testing sets.

Metric Category	Weighted				Macro			
	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score
Multiclass	0.9741	0.9826	0.9741	0.977	-	0.5712	0.6746	0.5992
Binary	0.9894	0.9266	0.9947	0.9594	-	-	-	-

Table 2: Trained Scores of the MoE Model

5.1 Optimization Dynamics

The training history demonstrates the effectiveness of the Cosine Annealing learning rate schedule combined with early stopping. As illustrated in Figure 1, the training loss descends logarithmically, beginning near 0.26 and smoothly approaching an asymptote near 0.16. The validation metrics follow this descent closely, maintaining an exceptionally stable validation accuracy above 0.98. However, heteroscedastic spiking is visible in the validation loss during the final epochs, indicating aggressive weight adjustments as the optimizer explores narrower local minima.

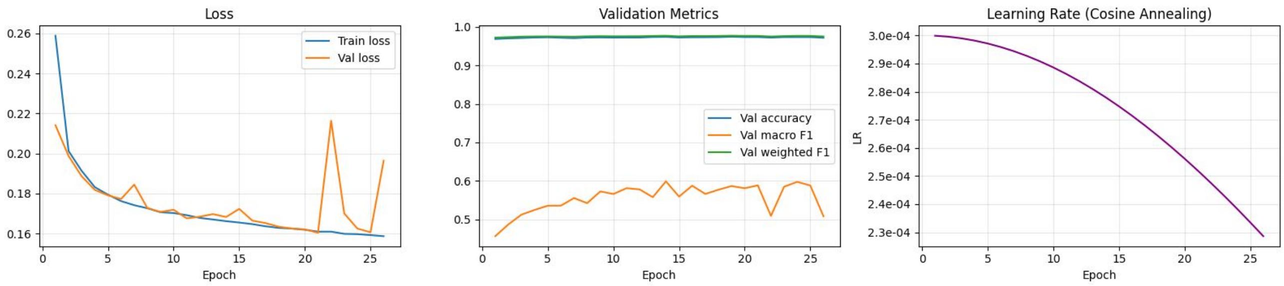

Figure 1: Training History

5.2 Multiclass Localization Efficacy

An analysis of the cross-class classification performance reveals extreme variability in the model's ability to localize specific attack categories. Figure 2 demonstrates that the model achieves near-perfection (0.99 F1-score) in identifying "Generic" and "Normal" traffic, driven by their massive representation in the training data geometry. Shellcode and Reconnaissance are also detected with high fidelity. However, the system struggles significantly with heavily obfuscated, minority payloads such as "Analysis" and "Backdoor".

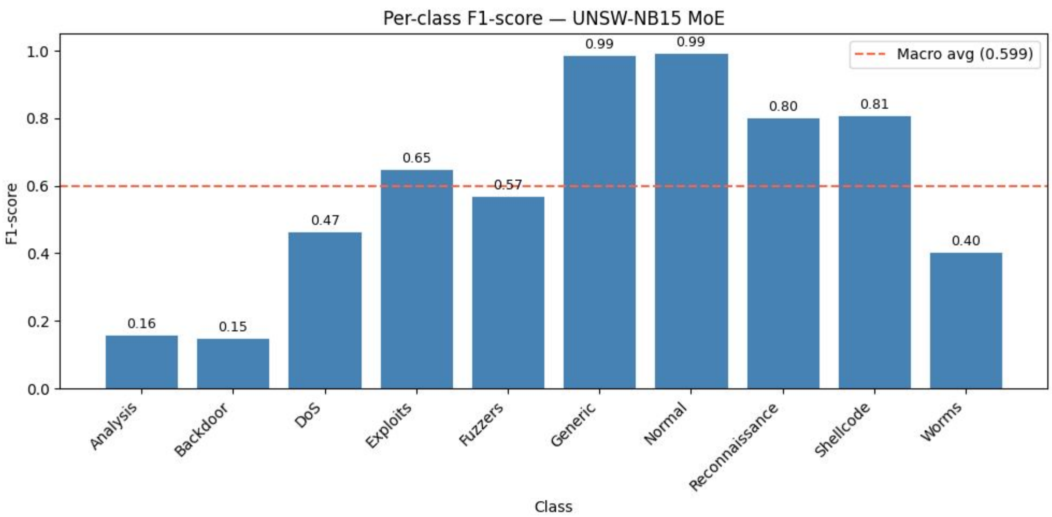

Figure 2: Per-class F1-score chart

5.3 Binary Threat Detection

Despite the localized struggles in the multiclass domain, the discriminative power of the model when reduced to a binary classification task (Normal vs. Attack) is exceptional. As shown in Figure 3, out of the evaluated flows, the model correctly identified 329,017 True Normal instances and 47,938 True Attack instances. Crucially, False Negatives were suppressed to an exceptionally low 255 instances.

5.4 One-vs-Rest ROC Analysis

To further untangle the multiclass performance, an evaluation of the One-vs-Rest (OvR) ROC Curves was conducted as shown in Figure 4, yielding an impressive Mean AUC of 0.996. The analysis reveals that the Generic, Normal, and Shellcode classes achieve an AUC of exactly 1.000. Notably, while the F1-scores for Backdoor and Analysis were critically low, their AUC scores remain exceptionally high. This striking discrepancy between F1 and AUC highlights that the model is successfully learning discriminative embeddings for minority classes; the errors primarily stem from calibration issues and thresholding within the flat, highly imbalanced probability space rather than an inherent inability to separate the classes.

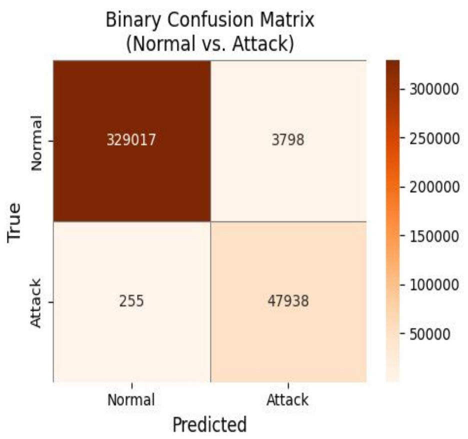

Figure 3: Binary Confusion Matrix

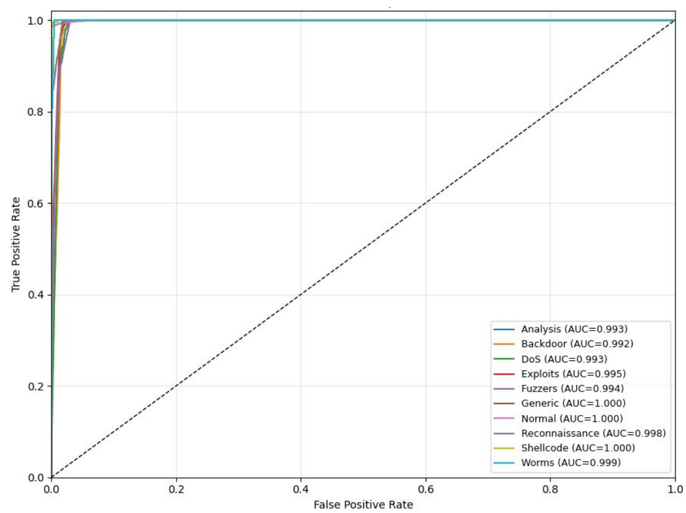

Figure 4: ROC Curves

5.4 Real-Time Stream Inference

To simulate a deployment environment, the model was tested through a Network Intrusion Stream (NIS) Inference dashboard as shown in Figure 5. Operating at a cadence of one flow every 3 seconds, the stream inference engine maintained a perfect running accuracy and binary precision. The continuous probability vectors output by the MoE gating network demonstrate the system's ability to translate complex anomaly detections into actionable, real-time metrics.

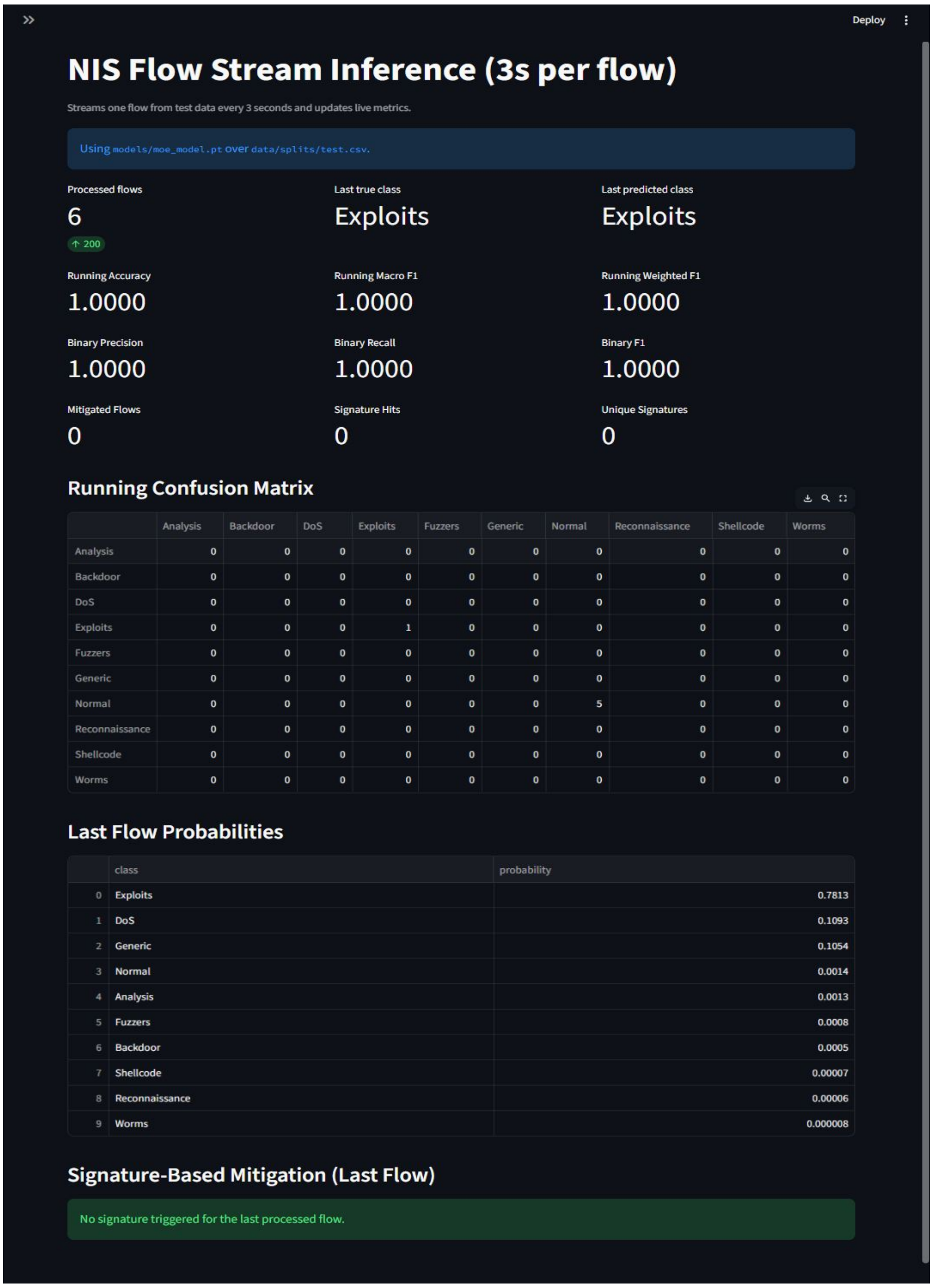

Figure 5: Streamlit UI

6. Discussion

The visual and empirical artifacts from the Main MoE pipeline reveal a stark dichotomy between the model’s feature extraction capabilities and its operational thresholding. The training history charts illustrate successful optimization, with training loss converging smoothly to 0.16. While validation accuracy remains exceptionally stable above 0.98, the persistent gap between validation accuracy and macro F1 (oscillating between 0.50–0.60) exposes the network's struggle with extreme class imbalance.

This limitation is explicitly quantified in the per-class F1-score distribution. The model achieves near-perfection (0.99) on majority classes like Generic and Normal, but completely fails on stealthy minority payloads like Analysis (0.16) and Backdoor (0.15). However, analyzing the One-vs-Rest ROC curves fundamentally reframes this failure: despite low F1 scores, the AUC for Analysis (0.993) and Backdoor (0.992) remains exceptionally high, yielding a mean AUC of 0.996. This proves the MoE experts successfully learned to separate the high-dimensional manifolds of these minority attacks; the low F1 scores are symptomatic of poor probability calibration and thresholding within a flat multiclass space, rather than an inherent failure of representational capacity.

Conversely, when reduced to a binary anomaly detection task, the engine excels. The binary confusion matrix shows only 255 false negatives out of over 370,000 flows, heavily prioritizing zero-trust threat containment. Finally, the NIS Flow Stream Inference dashboard confirms the model’s deployment viability. By successfully processing live flows and outputting continuous probability vectors (e.g., 0.7813 for Exploits), the system effectively bridges the gap between theoretical anomaly localization and real-time, signature-based threat mitigation.

7. Conclusion

A soft-gated Mixture-of-Experts neural network for multiclass network intrusion detection on the UNSW-NB15 dataset is presented. Three structurally heterogeneous experts — differing in depth, width, and regularization — are combined via a learned input-conditional gating network. Our square-root balanced class-weighting strategy addresses severe class imbalance without synthetic oversampling. Training is stabilized using cosine annealing and early stopping based on macro F1-score. Dual evaluation at multiclass and binary levels provides both technical completeness and operational relevance.

The key insight driving this work is that heterogeneous expert diversity — not simply the number of experts — is what enables the gating network to specialize meaningfully across nine structurally dissimilar attack families.

Future work will focus on: (i) ablation studies comparing individual experts to the full ensemble; (ii) visualization of per-class gating weights to verify expert specialization; (iii) extension to zero-day detection using held-out attack families; and (iv) evaluation on CIC-IDS2017/2018 for cross-dataset generalization.

8. References

[1] N. Moustafa and J. Slay, "UNSW-NB15: a comprehensive data set for network intrusion detection systems," in Proc. Military Communications and Information Systems Conference (MilCIS), IEEE, 2015.

[2] A. R. Sonule, M. Kalla, A. Jain, and D. S. Chouhan, "UNSW-NB15 dataset and machine learning based intrusion detection systems," Int. J. Engineering and Advanced Technology, vol. 9, 2020.

[3] R. A. Jacobs, M. I. Jordan, S. J. Nowlan, and G. E. Hinton, "Adaptive mixtures of local experts," Neural Computation, vol. 3, no. 1, pp. 79–87, 1991.

[4] N. Shazeer et al., "Outrageously large neural networks: The sparsely-gated mixture-of-experts layer," in Proc. ICLR, 2017.

[5] Zoghi, G. et al., "Building an intrusion detection system on UNSW-NB15: Reducing the margin of error to deal with data overlap and imbalance," Concurrency and Computation: Practice and Experience, vol. 36, no. 25, 2024.

[6] Abdelkhalek, A. and Mashaly, M., "Addressing the class imbalance problem in network intrusion detection systems using data resampling and deep learning," J. Supercomputing, vol. 79, pp. 10611–10644, 2023.

[7] S. Choudhary and N. Kesswani, "Analysis of KDD-Cup'99, NSL-KDD and UNSW-NB15 datasets using deep learning in IoT," Procedia Computer Science, vol. 167, pp. 1561–1573, 2020.

[8] H. Tariq et al., "A novel ensemble of deep learning approach for cybersecurity intrusion detection with explainable artificial intelligence," Applied Sciences, vol. 15, no. 14, 2025.

[9] B. S. Alotaibi et al., "Enhanced intrusion detection systems performance with UNSW-NB15 data analysis," Algorithms, vol. 17, no. 2, 2024.

[10] N. Shafiq et al., "A hybrid feature weighted attention based deep learning approach for an intrusion detection system using the random forest algorithm," PLOS ONE, 2024.

[11] M. Zolanvari et al., "Intrusion detection in IoT using supervised machine learning based on application and transport layer features using UNSW-NB15," EURASIP J. Wireless Communications and Networking, 2021.

[12] Agrafiotis, I. et al., "Convolutional neural networks and mixture of experts for intrusion detection in 5G networks and beyond," Frontiers in Communications and Networks, 2025.

[13] Z. Ahmad et al., "Network intrusion detection system: a systematic study of machine learning and deep learning approaches," Transactions on Emerging Telecommunications Technologies, vol. 32, 2021.

[14] W. Chen et al., "A survey on imbalanced learning: latest research, applications and future directions," Artificial Intelligence Review, vol. 57, 2024.